

- (c) Use your graph to solve the equation $\frac{1}{5}x^3 - \frac{4}{5}x^2 - 2 = 0$.

Answer $x =$ _____ [2]

- (d) By drawing a tangent, find the gradient of the curve at the point (4, 0).

Answer _____ [2]

- (e) (i) On the same axes, draw the line $y = 4 - 2x$ for $0 \leq x \leq 5$. [1]

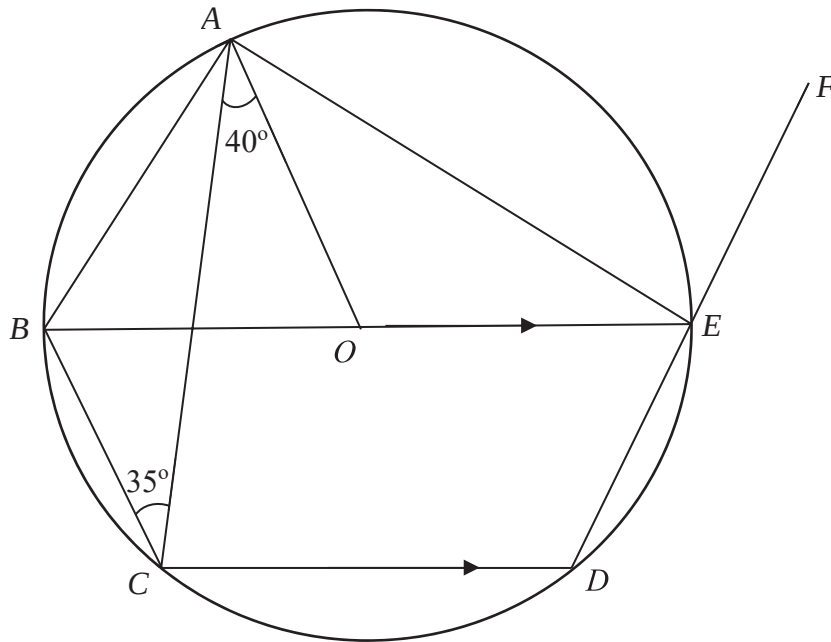
- (ii) Write down the x -coordinate of the point where this line intersects the curve.

Answer _____ [1]

- (iii) This value of x is a solution of the equation $x^3 - 4x^2 + Ax + B = 0$.
Find the value of A and of B .

Answer $A =$ _____, $B =$ _____ [3]

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A, B, C, D and E are points on the circle, with centre O .
 The diameter of the circle, BE , is parallel to CD .
 Angle $OAC = 40^\circ$ and angle $ACB = 35^\circ$.
 DEF is a straight line.

Find, giving reasons for each answer,

(a) angle OEA ,

Answer _____ [1]

(b) reflex angle AOB ,

Answer _____ [2]

(c) angle BAC ,

Answer _____ [2]

(d) angle FEB ,

Answer _____ [2]

(e) angle DOE .

Answer _____ [2]